

# Manual

## Upgrade: Detailed Planning Data for ERP EIS-EFD 8.1

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# 1 Upgrade: Detailed Planning Data for ERP

## Possible fields of application

The function package "modification of detailed scheduling data to ERP" enhances the ERP systems interface by the upload of detailed scheduling information.

## Implementation notes

The function package is used if you:

- use the detailed scheduling/shop floor scheduling module and
- would like to upload the detailed scheduling dates specified there to the ERP system

## Integration

The function package uses the data specified within the framework of detailed scheduling.

## Functions

- Upload of changes to the planning
  - Upload of detailed scheduling dates
  - Upload of the machine/workplace on which the OP was scheduled
  - Identification of conflicts

## 2 Setup of Data Record Structure

The data are transferred in the following structure. Within this structure the value of the SEGNAM field precisely defines the set-up of the user data structure in the SDATA field.

| Field name | Type | Length | Designation           | Data field and meaning                                                                                                                                                        |
|------------|------|--------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEGNAM*    | Char | 30     | Segment name          | This field is occupied by the writing system with the respective segment name. This precisely defines the set-up of the data record (SDATA field).<br>Example: HY72_AU_HD_001 |
| MANDT*     | Char | 3      | Client                | Reserved; fixed: '000'                                                                                                                                                        |
| DOCNUM*    | Char | 16     | IDOC number           | Serial number for the IDOCs<br>Reserved: fixed '0000000000000000'                                                                                                             |
| SEGNUM*    | Char | 6      | Segment number        | Reserved: fixed '000000'                                                                                                                                                      |
| PSEGNUM    | Char | 6      | Parent segment number | Reserved; fixed: '000000'                                                                                                                                                     |
| HLEVEL     | Char | 2      | Hierarchy level       | Reserved; fixed: '00'                                                                                                                                                         |
| SDATA      | Char | 1000   | User data             | This field contains the user data. The structure of this field is defined by the SEGNAM field.                                                                                |

### 3 Uploading data types: HYDRA --> ERP

| Type             | Description                                                                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CHAR x</b>    | Information is left-aligned for the data type CHAR; unnecessary places are filled with blanks.<br><br>Example: "ABCD "                                                                                                                                                                                                                                  |
| <b>NUM x</b>     | Numeric field of the length x without sign. Numbers are right-aligned; unnecessary places are filled with zeros.<br><br>Example: "00000002"                                                                                                                                                                                                             |
| <b>DEC_O x.y</b> | Numeric field of the length x and y decimal places. An algebraic sign is preceding the data field ("+" or "-"). Places that are not required are filled with zeros. There is NO DECIMAL SEPARATOR.<br><br>e.g. DEC_O 13,3: <ul style="list-style-type: none"> <li>• -1234567890,123 → -1234567890123</li> <li>• 234567890,3 → +0234567890300</li> </ul> |
| <b>DATE</b>      | The date is displayed in the YYYYMMDD format.<br><br>The field is filled with blanks (if it is not required).                                                                                                                                                                                                                                           |
| <b>TIME</b>      | The time is transferred in the HHMMSS format.<br><br>The field is populated with "000000".                                                                                                                                                                                                                                                              |

Generally, HYDRA always transfers a contiguous data structure. Data fields that are not used are filled with blanks. The following definitions apply if you use the file port:

Each data record included in the file has to be completed by 'CR' (U+000D) and 'LF' (U+000A) for Windows and 'LF' (U+000A) for Unix.

HYDRA expects the file to be in the UTF-8 format and HYDRA also uses this format for uploads. On request, the file transfer may also be performed in the file format that was used until MW 2.0.

## 4 Upload of Planning Changes

PPS specifications can be modified or infringed by assigning operations in the HYDRA shop floor scheduling module (HYDRA-HLS). These modifications can be uploaded to the PPS system.



Only the modifications in the graphical planning board of the HYDRA shop floor scheduling system (HLS) will be uploaded. Modifications made by the editing functions for operations or the order sequencing will not be taken into account.

The following modifications and/ or statuses will be confirmed/uploaded:

- Planning for a workplace in HLS
- Re-planning within a workplace in HLS
- Re-planning for a different workplace (workplace/ machine change) in HLS
- Deallocation of the operation in HLS

The customer will interpret the confirmations/uploads as well as the processing resulting for the PPS system.

The following optional parameters can be set at the order type using an upload indicator (HYDRA customizing) for MLE communication:

- Upload of all operations to the PPS system .  
(Default: only the operations generated by PPS will be uploaded)
- Only upload of delayed operations  
(default: inactive)
- Only upload of operations with different machine occupation/assignment  
(default: inactive)

The below specifications apply for the Idoc:

**Message type:** HY72ADRCK\_SC

**IDOC type:** HY72ADRCK\_SC

**Segment file name:** HY72ADRCK\_SCHEDULE

**File extension:** According to the configuration in the MLE communication  
(Logical systems > Outbound Configuration Fileport)  
Normally: ".dat"

| Field    | T     | L  | D | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-------|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ANR      | CHAR  | 40 |   | HYDRA order number = combined order/ operation number<br>The precise length that will be confirmed/uploaded depends on the length configuration of the order and/or operation in the HYDRA basic settings.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| RMNR     | CHAR  | 40 |   | Upload number (if available in HYDRA)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| AUART    | CHAR  | 5  |   | Order type of an order; according to HYDRA configuration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| DAT      | DATE  | 8  |   | Date of fixing (saving) of the reallocation in HLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| ZEI      | TIME  | 6  |   | Time of fixing (saving) of the reallocation in HLS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| SGR:GUTB | DEC_O | 13 | 3 | Target quantities in base quantity unit if stored to the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| SGE:B    | CHAR  | 3  |   | Base quantity unit if stored to the operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| SGR:GUTP | DEC_O | 13 | 3 | Target quantities in primary quantity unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| SGE:P    | CHAR  | 3  |   | Primary quantity unit: Primary unit of entry from the operation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| DATB     | DATE  | 8  |   | These data fields include the start time scheduled by HLS for this operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| ZEIB     | TIME  | 6  |   | These fields are not defined with the ACTION = "G".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| DATE     | DATE  | 8  |   | These data fields include the end time scheduled by HLS for this operation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ZEIE     | TIME  | 6  |   | These fields are not defined with the ACTION = "G".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| MNR      | CHAR  | 8  |   | Workplace, to which the operation was allocated.<br>This field is not defined with the ACTION = "G".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| MGRP     | CHAR  | 8  |   | ACTION = "M" or "U":<br>Group of the workplace, to which the operation was allocated.<br>ACTION = "G":<br>Group, to which the operation was deallocated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ART      | CHAR  | 1  |   | "K" = if a conflict is found when planning changes are uploaded for the operation (see KONFLIKT field), otherwise undefined.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| AKTION   | CHAR  | 2  |   | "M " - operation was allocated to workplace MNR. This may be an allocation from the pool of groups to a workplace as well as a reallocation of one workplace to another.<br>"U " - Operation was rescheduled within the workplace MNR.<br>"G " - Operation was deallocated to the pool of groups.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| KONFLIKT | CHAR  | 2  |   | Prio 1: "PE" - The end date of the operation exceeds the basic date of the order specified by the PPS system<br>Prio 2: "TE" - The operation exceeds the latest end date resulting from scheduling.<br>Prio 3: "PM" - Workplace is not the workplace transferred by the PPS system (during the first data transfer of the operation). Only possible if the operation was transferred by the PPS system.<br>PLEASE NOTE: Only the conflict of the highest priority will be uploaded.<br>Example: If a planned operation has exceeded the basic date of the order and if an operation was allocated to another scheduled workplace transferred by the PPS system, only the "PE" conflict will be uploaded due to scheduling. |



If the additional function to split operations is used in the HYDRA shop floor scheduling system (HLS-AGS license),



- split masters will not be uploaded since they are no longer planned after splitting;
- split operations will only be uploaded if the "Upload PPS operations only" option is NOT set in the order type, "confirmations" tab.

## 5 HYDRA Settings relevant for the Application

Proceed as described in the following to activate the interface for the upload of planning changes from the HYDRA Shop Floor Scheduling (HLS):

### Activating the logging

Edit the following entries in the [Logging configuration](#):

| Parameter name                              | Value                       |
|---------------------------------------------|-----------------------------|
| <b>For the scheduling of operations</b>     |                             |
| Object                                      | HLS                         |
| Action                                      | EINPLANEN                   |
| Logging of dialog data                      | Yes                         |
| Logging                                     | Entire object               |
| Comment                                     | No                          |
| Labeling                                    | No                          |
| Segment – for the upload to non-SAP systems | HY72ADRCK_SCHEDULE          |
| Segment – for the upload to SAP systems     | Z2HY72ADRCK_SCHEDULE000X000 |
| <b>For the replanning of operations</b>     |                             |
| Object                                      | HLS                         |
| Action                                      | UMPLANEN                    |
| Logging of dialog data                      | Yes                         |
| Logging                                     | Entire object               |
| Comment                                     | No                          |
| Labeling                                    | No                          |
| Segment – for the upload to non-SAP systems | HY72ADRCK_SCHEDULE          |
| Segment – for the upload to SAP systems     | Z2HY72ADRCK_SCHEDULE000X000 |
| <b>For the deallocation of operations</b>   |                             |
| Object                                      | HLS                         |
| Action                                      | AUSPLANEN                   |
| Logging of dialog data                      | Yes                         |
| Logging                                     | Entire object               |
| Comment                                     | No                          |
| Labeling                                    | No                          |
| Segment – for the upload to non-SAP systems | HY72ADRCK_SCHEDULE          |
| Segment – for the upload to SAP systems     | Z2HY72ADRCK_SCHEDULE000X000 |

## Activating the upload for the order type

If required, activate for the relevant [Order type](#) (s) > *Confirmation* > *Confirmation of planning changes in shop floor scheduling*:

| Parameter name                                   | Value                |
|--------------------------------------------------|----------------------|
| Upload PPS operations only                       | Activate if required |
| Upload delayed operations only                   | Activate if required |
| Upload operations with deviating assignment only | Activate if required |

### Upload PPS operations only

Upload PPS operations only

If you activate this option, the operations created in HYDRA are not uploaded.

### Upload delayed operations only

Upload delayed operations only

If you activate this option, only delayed operations are uploaded.

### Upload operations with deviating assignment only

Upload operations with deviating assignment only

If you activate this option, only operations are uploaded that have been planned for another workplace than the workplace initially transferred.

You can use the option *Upload PPS operations only* no matter whether the other options are enabled or not. But you can only enable one of the two options *Upload delayed operations only* and *Upload operations with deviating assignment only*.

## Activating the upload in the scheduler

Use the [HYDRA Scheduler](#) to plan jobs for the outbound processing:

| Parameter name                                                                     | Value                                                                |
|------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <b><i>Transfer of uploads to the <a href="#">MLE outbound transactions</a></i></b> |                                                                      |
| Product key                                                                        |                                                                      |
| License key                                                                        |                                                                      |
| Command (Windows):                                                                 | sh.exe ./myerprck.scr /LOGGING<br>/LOGGING_SEGNAM=HY72ADRCK_SCHEDULE |
| Command (Unix):                                                                    | ./myerprck.scr /LOGGING<br>/LOGGING_SEGNAM=HY72ADRCK_SCHEDULE        |
| Comment:                                                                           | EIS-EFD: Upload of planning changes<br>-> MLE outbound transactions  |

| Parameter name                                                                                      | Value                                                                       |
|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Interval                                                                                            | 5                                                                           |
| <b>Upload of confirmations from the <a href="#">MLE outbound transactions</a> to the ERP system</b> |                                                                             |
| Product key                                                                                         |                                                                             |
| License key                                                                                         |                                                                             |
| Command (Windows) for the upload to a non-SAP system:                                               | sh.exe ./hysapupl.scr<br>/UPLSEGNAM=HY72ADRCK_SCHEDULE                      |
| Command (Windows) for the upload to an SAP system:                                                  | sh.exe ./hysapupl.scr<br>/UPLSEGNAM=Z2HY72ADRCK_SCHEDULE000X000             |
| Command (Unix) for the upload to a non-SAP system:                                                  | ./hysapupl.scr /UPLSEGNAM=HY72ADRCK_SCHEDULE                                |
| Command (Unix) for the upload to an SAP system:                                                     | ./hysapupl.scr<br>/UPLSEGNAM=Z2HY72ADRCK_SCHEDULE000X000                    |
| Comment:                                                                                            | EIS-EFD: Upload of planning changes MLE outbound transactions -> ERP system |
| Interval                                                                                            | 5                                                                           |

## Editing the HYDRA distribution model – output for non-SAP systems

Use the [HYDRA distribution model](#) to edit an entry for the HYDRA outbound processing:

| Parameter name                | Value                                |
|-------------------------------|--------------------------------------|
| <b>To upload time tickets</b> |                                      |
| Message type                  | HY72ADRCK_SC                         |
| Description                   | EIS-EFD – upload of planning changes |
| IDoc type                     | HY72ADRCK_SC                         |
| Retention period              | 10                                   |
| Log. target system            | Created logical system               |
| Segment name 1                | HY72ADRCK_SCHEDULE                   |

## Editing the HYDRA distribution model – output for SAP systems

Use the [HYDRA distribution model](#) to edit an entry for the HYDRA outbound processing:

| Parameter name                | Value                                |
|-------------------------------|--------------------------------------|
| <b>To upload time tickets</b> |                                      |
| Message type                  | ZHY72ADRCK_SC                        |
| Description                   | EIS-EFD – upload of planning changes |
| IDoc type                     | ZHY72ADRCK_SC01                      |
| Retention period              | 10                                   |

| Parameter name     | Value                       |
|--------------------|-----------------------------|
| Log. target system | Created logical system      |
| Segment name 1     | Z2HY72ADRCK_SCHEDULE000X000 |

## 6 Test Files

### Overview

Attached to this documentation, you will find test files for the interface EIS-EFD. The attachment is only available, if the documentation is in PDF format.

The documentation [Open PDF attachments](#) describes how to call the attached test files.

The following test files are attached to the PDF document:

| File                   | Type                | Comment                                                                 |
|------------------------|---------------------|-------------------------------------------------------------------------|
| HY72ADRCK_SCHEDULE.dat | Outbound processing | Sample file for the upload of planning changes in HYDRA standard format |